

Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Integrals

1.1 Medium Integral

Lambert W Function Integral

Evaluate the definite integral:

$$\int_0^e \frac{W(x)}{x(1+W(x))(1+W(x)^2)} dx$$

where $W(x)e^{W(x)} = x$.

Solution: We utilize the substitution based on the definition of the Lambert W function. Let $u = W(x)$. By definition, $x = ue^u$. Differentiating both sides with respect to u , we obtain the differential:

$$dx = (e^u + ue^u) du = e^u(1+u) du$$

Next, we determine the new limits of integration. When $x = 0$, $ue^u = 0 \implies u = 0$. When $x = e$, we have $ue^u = e$, which by inspection yields $u = 1$. Now we substitute u , x , and dx into the original integral:

$$I = \int_0^1 \frac{u}{(ue^u)(1+u)(1+u^2)} e^u(1+u) du$$

Notice that the terms e^u , $(1+u)$, and u in the numerator and denominator cancel out beautifully:

$$I = \int_0^1 \frac{1}{1+u^2} du$$

This is a standard integral evaluating to the arctangent function:

$$I = [\arctan(u)]_0^1 = \arctan(1) - \arctan(0)$$

$$I = \frac{\pi}{4}$$

1.2 Hard Integral

Product of Inverse Cotangents

Evaluate the definite integral:

$$\int_0^\infty \cot^{-1}(4x) \cot^{-1}(8x) dx$$

Solution: Consider the general integral $I(a, b) = \int_0^\infty (ax)(bx) dx$. We can apply integration by parts. Let $u = (ax)(bx)$ and $dv = dx$, yielding $v = x$. Using the

boundary limit $\lim_{x \rightarrow \infty} x(ax)(bx) = 0$, the boundary term vanishes, giving:

$$I(a, b) = \int_0^\infty \frac{ax}{1+a^2x^2}(bx) dx + \int_0^\infty \frac{bx}{1+b^2x^2}(ax) dx$$

Let's analyze the function $F(c) = \int_0^\infty \frac{x}{1+x^2}(cx) dx$. Differentiating $F(c)$ with respect to c using Leibniz's rule:

$$F'(c) = \int_0^\infty \frac{-x^2}{(1+x^2)(1+c^2x^2)} dx$$

Using partial fractions on the integrand:

$$\frac{-x^2}{(1+x^2)(1+c^2x^2)} = \frac{1}{c^2-1} \left(\frac{1}{1+c^2x^2} - \frac{1}{1+x^2} \right)$$

Integrating this from 0 to ∞ yields:

$$F'(c) = \frac{1}{c^2-1} \left(\frac{\pi}{2c} - \frac{\pi}{2} \right) = \frac{\pi}{2(c^2-1)} \left(\frac{1-c}{c} \right) = -\frac{\pi}{2c(c+1)}$$

Integrating $F'(c)$ with respect to c to recover $F(c)$:

$$F(c) = -\frac{\pi}{2} \int \left(\frac{1}{c} - \frac{1}{c+1} \right) dc = -\frac{\pi}{2} (\ln c - \ln(c+1)) + C = \frac{\pi}{2} \ln \left(1 + \frac{1}{c} \right) + C$$

Since $(cx) \rightarrow 0$ as $c \rightarrow \infty$, $F(\infty) = 0$, implying $C = 0$. Now, returning to $I(a, b)$, by substituting $u = ax$ into the first term and $u = bx$ into the second, we express them via F :

$$I(a, b) = \frac{1}{a} F\left(\frac{b}{a}\right) + \frac{1}{b} F\left(\frac{a}{b}\right) = \frac{\pi}{2a} \ln \left(1 + \frac{a}{b} \right) + \frac{\pi}{2b} \ln \left(1 + \frac{b}{a} \right)$$

Given $a = 4$ and $b = 8$:

$$I(4, 8) = \frac{\pi}{8} \ln \left(1 + \frac{4}{8} \right) + \frac{\pi}{16} \ln \left(1 + \frac{8}{4} \right)$$

$$I(4, 8) = \frac{\pi}{8} \ln \left(\frac{3}{2} \right) + \frac{\pi}{16} \ln(3) = \frac{\pi}{16} \left(2 \ln \left(\frac{3}{2} \right) + \ln(3) \right) = \frac{\pi}{16} \ln \left(\frac{9}{4} \cdot 3 \right)$$

$$I = \frac{\pi}{16} \ln \left(\frac{27}{4} \right)$$

Differentials

2.1 Medium Differential

First-Order Non-Linear ODE

Consider the differential equation:

$$\frac{dy}{dx} = \frac{1}{x + y^2}$$

Given that the solution curve passes through the point $(-2, 0)$, find the exact value of x when $y = 3$.

Solution: The given differential equation is non-linear in y , but can be made linear by treating x as a function of y . Take the reciprocal of both sides:

$$\frac{dx}{dy} = x + y^2 \implies \frac{dx}{dy} - x = y^2$$

This is a standard first-order linear differential equation in the form $\frac{dx}{dy} + P(y)x = Q(y)$. The integrating factor (IF) is:

$$\mu(y) = e^{\int -1 dy} = e^{-y}$$

Multiplying the ODE by the IF gives:

$$e^{-y} \frac{dx}{dy} - e^{-y} x = y^2 e^{-y} \implies \frac{d}{dy}(x e^{-y}) = y^2 e^{-y}$$

Integrating both sides with respect to y :

$$x e^{-y} = \int y^2 e^{-y} dy$$

Use integration by parts twice on the right-hand side. Let $u = y^2$, $dv = e^{-y} dy$. Then $du = 2y dy$, $v = -e^{-y}$.

$$\int y^2 e^{-y} dy = -y^2 e^{-y} + 2 \int y e^{-y} dy = -y^2 e^{-y} - 2y e^{-y} - 2e^{-y} + C$$

Dividing through by e^{-y} , we obtain the general solution:

$$x(y) = -y^2 - 2y - 2 + C e^y$$

Apply the initial condition $(-2, 0)$, which means $x = -2$ when $y = 0$:

$$-2 = -(0)^2 - 2(0) - 2 + C e^0 \implies -2 = -2 + C \implies C = 0$$

The specific solution is $x = -y^2 - 2y - 2$. To find x when $y = 3$:

$$x(3) = -(3)^2 - 2(3) - 2 = -9 - 6 - 2 = -17$$

2.2 Hard Differential

Isogonal Trajectories

Let ρ be the family of all straight lines passing through the origin. Find a curve χ such that the tangent of the angle between ρ and χ is e . Given $y(5) = 5$, evaluate the constant of integration C in the form:

$$\frac{A}{e} - \ln(\sqrt{50}) = C$$

Find the value of $|A|$.

Solution: Let the lines ρ passing through the origin be represented in polar coordinates simply as radial lines. The angle between the radius vector (line ρ) and the tangent to the curve χ is denoted by α . From differential geometry in polar coordinates, we know:

$$\tan \alpha = r \frac{d\theta}{dr}$$

The problem states that the tangent of the angle between ρ and χ is e . Therefore:

$$r \frac{d\theta}{dr} = \pm e$$

Separating the variables yields:

$$\frac{dr}{r} = \pm \frac{1}{e} d\theta$$

Integrating both sides gives the family of logarithmic spirals:

$$\ln r = \pm \frac{\theta}{e} - C$$

(We use $-C$ as an arbitrary constant to match the target equation format). We are given an initial condition in Cartesian coordinates: $y(5) = 5$. The point is $(5, 5)$. Convert this to polar coordinates:

$$r = \sqrt{5^2 + 5^2} = \sqrt{50}$$

$$\theta = \arctan\left(\frac{5}{5}\right) = \frac{\pi}{4}$$

Substitute these into the general solution (choosing the positive branch for this case, which results in the matching algebraic structure):

$$\ln(\sqrt{50}) = \frac{\pi/4}{e} - C \implies C = \frac{\pi/4}{e} - \ln(\sqrt{50})$$

Comparing this strictly to the specified form $C = \frac{A}{e} - \ln(\sqrt{50})$, we deduce:

$$A = \frac{\pi}{4}$$

Thus, taking the absolute value, the result is:

$$|A| = \frac{\pi}{4}$$

Derivatives

3.1 Medium Derivative

Implicit Functional Derivative

Let f be differentiable satisfying $f(x + f(x)) = x$. Find $f'(0)$.

Solution: First, we determine the value of $f(0)$. Let $f(0) = c$. Substituting $x = 0$ into the functional equation gives:

$$f(0 + f(0)) = 0 \implies f(c) = 0$$

Next, substitute $x = c$ into the original functional equation:

$$f(c + f(c)) = c$$

Since we just established that $f(c) = 0$, this simplifies to:

$$f(c + 0) = c \implies f(c) = c$$

However, we already know $f(c) = 0$. Therefore, $c = 0$, which strictly proves that $f(0) = 0$.

Now, differentiate both sides of the original functional equation with respect to x using the chain rule:

$$f'(x + f(x)) \cdot (1 + f'(x)) = 1$$

Evaluate this derivative expression at $x = 0$:

$$f'(0 + f(0)) \cdot (1 + f'(0)) = 1$$

Substitute our finding $f(0) = 0$:

$$f'(0) \cdot (1 + f'(0)) = 1$$

Let $k = f'(0)$. This forms a standard quadratic equation:

$$k(1 + k) = 1 \implies k^2 + k - 1 = 0$$

Applying the quadratic formula yields the exact values for the derivative:

$$k = \frac{-1 \pm \sqrt{1 - 4(1)(-1)}}{2} = \frac{-1 \pm \sqrt{5}}{2}$$

Therefore, the possible values for $f'(0)$ are $\frac{-1+\sqrt{5}}{2}$ and $\frac{-1-\sqrt{5}}{2}$.

3.2 Hard Derivative

Symmetric Implicit Derivative

Let a family of curves in the first quadrant be defined by

$$\ln(x^y + y^x) = \sum_{n=1}^{\infty} \frac{(x^2 + y^2)^{-n}}{n}$$

Let $P = (x_0, y_0)$ be the unique point on this curve where $x_0 = y_0$. Find the value of $\frac{dy}{dx}$ at the point P .

Solution: First, we recognize the infinite series on the right side as the Taylor series expansion for the natural logarithm:

$$-\ln(1 - z) = \sum_{n=1}^{\infty} \frac{z^n}{n} \quad \text{for } |z| < 1$$

Let $z = \frac{1}{x^2 + y^2}$. Then the right side simplifies to:

$$-\ln\left(1 - \frac{1}{x^2 + y^2}\right) = \ln\left(\frac{x^2 + y^2}{x^2 + y^2 - 1}\right)$$

The full curve equation is then algebraically defined as:

$$x^y + y^x = \frac{x^2 + y^2}{x^2 + y^2 - 1}$$

Let this equation be rewritten as $F(x, y) = 0$, where:

$$F(x, y) = x^y + y^x - \frac{x^2 + y^2}{x^2 + y^2 - 1}$$

Notice that this function is completely symmetric with respect to variables x and y , meaning $F(x, y) = F(y, x)$. Using implicit differentiation, the derivative $\frac{dy}{dx}$ is given by:

$$\frac{dy}{dx} = -\frac{F_x}{F_y}$$

where F_x and F_y are the partial derivatives of F with respect to x and y , respectively. Since F is completely symmetric, evaluating the partial derivatives at any point where $x = y$ (the point P) guarantees that $F_x(x_0, x_0) = F_y(x_0, x_0)$. Provided that $F_x(x_0, x_0) \neq 0$, the derivative simplifies purely analytically:

$$\left.\frac{dy}{dx}\right|_P = -\frac{F_x(x_0, x_0)}{F_x(x_0, x_0)} = -1$$

Physics & Engineering

4.1 Mechanics

Kinematics of a Basketball Pass

A basketball is thrown with an initial speed $v_0 = 12.5 \text{ m s}^{-1}$ at an angle $\theta = 53^\circ$ above the horizontal. A teammate starts $d = 7 \text{ m}$ away and runs to catch the ball at the same height it was released. His acceleration is $a = Kt$. Find the constant K if he catches the ball. Take $g = 10 \text{ m s}^{-2}$, $\sin(53^\circ) = 0.8$, and $\cos(53^\circ) = 0.6$.

Solution: First, calculate the horizontal and vertical components of the initial velocity:

$$v_{0x} = v_0 \cos(53^\circ) = 12.5(0.6) = 7.5 \text{ m s}^{-1}$$

$$v_{0y} = v_0 \sin(53^\circ) = 12.5(0.8) = 10.0 \text{ m s}^{-1}$$

The time of flight T is determined by the vertical motion since the ball is caught at the same release height ($\Delta y = 0$):

$$\Delta y = v_{0y}T - \frac{1}{2}gT^2 = 0 \implies T = \frac{2v_{0y}}{g} = \frac{2(10)}{10} = 2 \text{ s}$$

The total horizontal distance (range) the ball travels is:

$$R = v_{0x}T = 7.5(2) = 15 \text{ m}$$

The teammate starts 7 m away from the thrower and runs in the direction of the ball. Therefore, the distance he must cover to catch it is $15 - 7 = 8 \text{ m}$. The teammate's acceleration is given by $a(t) = Kt$. Integrating to find velocity (assuming starting from rest, $v(0) = 0$):

$$v(t) = \int Kt dt = \frac{1}{2}Kt^2$$

Integrating again to find distance covered $x(t)$:

$$x(t) = \int \frac{1}{2}Kt^2 dt = \frac{1}{6}Kt^3$$

At $t = T = 2 \text{ s}$, the teammate covers 8 m:

$$8 = \frac{1}{6}K(2)^3 = \frac{8}{6}K = \frac{4}{3}K$$

Solving for K :

$$K = \frac{8 \times 3}{4} = 6 \text{ m s}^{-3}$$

4.2 Quantum Mechanics

Compton Scattering and Magnetic Deflection

An incident photon, whose energy is $\eta = 2.0$ times the rest energy of an electron, undergoes a head-on Compton collision with a stationary free electron. Following the collision, the recoil electron enters a uniform magnetic field of $B = 0.27$ T, moving perpendicularly to the magnetic field lines. Calculate the radius of curvature of the electron's trajectory in cm. Assume $m_e = 9.0 \times 10^{-31}$ kg, $e = 1.6 \times 10^{-19}$ C, and $c = 3.0 \times 10^8$ m s⁻¹.

Solution: Let $E_0 = m_e c^2$ be the rest energy of the electron. Incident photon energy is $E_\gamma = 2.0E_0$. For a head-on collision, the photon scatters backwards ($\theta = 180^\circ$). The Compton shift equation for energy is:

$$\frac{1}{E'_\gamma} - \frac{1}{E_\gamma} = \frac{1 - \cos(180^\circ)}{m_e c^2} = \frac{2}{E_0}$$

Substituting $E_\gamma = 2.0E_0$:

$$\frac{1}{E'_\gamma} - \frac{1}{2.0E_0} = \frac{2}{E_0} \implies \frac{1}{E'_\gamma} = \frac{2.5}{E_0} \implies E'_\gamma = 0.4E_0$$

By conservation of energy, the kinetic energy K_e transferred to the electron is:

$$K_e = E_\gamma - E'_\gamma = 2.0E_0 - 0.4E_0 = 1.6E_0$$

The total relativistic energy of the recoiling electron is $E_e = K_e + E_0 = 2.6E_0$. Using the energy-momentum relation $E_e^2 = (p_e c)^2 + E_0^2$, we solve for the momentum p_e :

$$(2.6E_0)^2 = (p_e c)^2 + E_0^2 \implies (p_e c)^2 = (6.76 - 1)E_0^2 = 5.76E_0^2$$

$$p_e c = \sqrt{5.76}E_0 = 2.4E_0 = 2.4m_e c^2$$

Therefore, the momentum is $p_e = 2.4m_e c$. When the electron enters the magnetic field, the radius of curvature R is given by $p_e = eBR$, yielding:

$$R = \frac{p_e}{eB} = \frac{2.4m_e c}{eB}$$

Substitute the provided values:

$$R = \frac{2.4 \times (9.0 \times 10^{-31}) \times (3.0 \times 10^8)}{(1.6 \times 10^{-19}) \times 0.27}$$

$$R = \frac{64.8 \times 10^{-23}}{0.432 \times 10^{-18}} = \frac{64.8}{43.2} \times 10^{-2} \text{ m} = 1.5 \times 10^{-2} \text{ m}$$

Converting to centimeters:

$$R = 1.5 \text{ cm}$$

4.3 Electromagnetism

Maximum Current in an LC Circuit

In an ideal LC circuit, an inductor of inductance $L = 50 \text{ mH}$ is connected to a fully charged capacitor of capacitance $C = 20 \mu\text{F}$. If the maximum initial charge on the capacitor is $Q_0 = 100 \mu\text{C}$, calculate the maximum current that will flow through the inductor during the oscillations in milliamperes (mA).

Solution: In an ideal LC circuit, the charge and current undergo simple harmonic oscillation. The angular frequency ω of the oscillation is:

$$\omega = \frac{1}{\sqrt{LC}}$$

Substitute the given values for L and C :

$$L = 50 \text{ mH} = 50 \times 10^{-3} \text{ H} = 0.05 \text{ H}$$

$$C = 20 \mu\text{F} = 20 \times 10^{-6} \text{ F}$$

$$\omega = \frac{1}{\sqrt{0.05 \times 20 \times 10^{-6}}} = \frac{1}{\sqrt{10^{-6}}} = 10^3 \text{ rad/s}$$

The maximum charge is $Q_0 = 100 \mu\text{C} = 100 \times 10^{-6} \text{ C}$. The relationship between maximum current $I_{\max}$ and maximum charge Q_0 in an LC circuit is given by the derivative amplitude:

$$I_{\max} = \omega Q_0$$

Evaluating this product:

$$I_{\max} = (10^3 \text{ rad/s}) \times (100 \times 10^{-6} \text{ C}) = 100 \times 10^{-3} \text{ A}$$

Converting amperes to milliamperes:

$$I_{\max} = 100 \text{ mA}$$